

3.2 SYSTEM DESIGN

The system design is a critical part of the project as it outlines how the entire disease prediction system works from data collection to disease identification. This chapter elaborates on the architecture, components, and flow of the system. The block diagram has been included to depict the process flow clearly.

3.2.1 Overview of System Design

The system is structured to efficiently process real-time image data of banana plant leaves, apply preprocessing techniques, augment data for robustness, and train a deep learning models for disease prediction. The main stages of the system design are:

- 1. Dataset Collection**
- 2. Image Preprocessing**
- 3. Data Augmentation**
- 4. Model Selection**
- 5. Model Training**
- 6. Model Evaluation**
- 7. Disease Prediction Output**

Each component is interconnected and works together to accomplish accurate disease classification.

1. Dataset Collection

- Source: The dataset used in this project consists of real-time banana leaf images captured directly from farms by collaborating with local farmers.
- Categories: The images are categorized into:
 - Black Sigatoka
 - Yellow Sigatoka
 - Potassium Deficiency
 - Panama Disease
 - Healthy
- Storage Structure: The dataset is organized into:

```
BananaLeafDataset/  
├── train/  
│   ├── black_sigatoka/  
│   ├── yellow_sigatoka/  
│   ├── healthy/  
│   ├── potassium_deficiency/  
│   └── panama_disease/  
├── val/  
│   └── same structure as train  
└── test/  
    └── same structure as train
```

Each sub-folder contains the respective classes.

- Split:
 - Train: 70%
 - Validation: 15%
 - Test: 15%

This structure allows for streamlined loading during model training and evaluation.

2. Image Preprocessing

Preprocessing ensures that the model receives consistent and clean data.

Key preprocessing steps include:

- Resizing: Images are resized to a fixed dimension (e.g., 224x224) required by ResNeXt.
- Normalization: Pixel values are normalized (scaled between 0 and 1 or mean-centered) to aid model convergence.
- Color Channel Conversion: Ensure consistent RGB format.
- Noise Removal: Techniques like Gaussian blurring are optionally used to reduce irrelevant noise.

These steps make the raw images suitable for training the deep learning model.

3. Data Augmentation

To avoid overfitting and improve model generalization, augmentation is performed. It synthetically increases the size of the dataset and introduces variability.

Common augmentation techniques used:

- Rotation (e.g., ± 20 degrees)
- Flipping (horizontal/vertical)
- Zooming
- Shearing
- Brightness/contrast adjustments

Libraries like `torchvision.transforms` or `imgaug` are utilized in Google Colab.

4. Model Selection

Several deep learning models were evaluated.

Model Comparison Considered:

- EfficientNet
- MobileNetV2
- DenseNet
- InceptionV3
- CNN (Vanilla)

5. Model Training

- Platform: Google Colab (GPU-enabled)
- Framework: PyTorch with `timm` library for model loading.
- Optimizer: Adam or SGD with weight decay.

- Loss Function: Cross-Entropy Loss (as it's a multi-class classification problem).
- Learning Rate: Tuned using scheduler (StepLR or ReduceLROnPlateau).
- Batch Size: 32
- Epochs: 20–30 depending on early stopping.

Logs and metrics (accuracy/loss curves) are tracked for each epoch to evaluate convergence.

6. Model Evaluation

Evaluation is based on validation and test datasets. Key performance metrics:

- Accuracy
- Precision, Recall, F1-Score
- Confusion Matrix
- ROC-AUC Score (for multi-class problems)

Misclassified samples are analyzed to understand edge cases and improve performance.

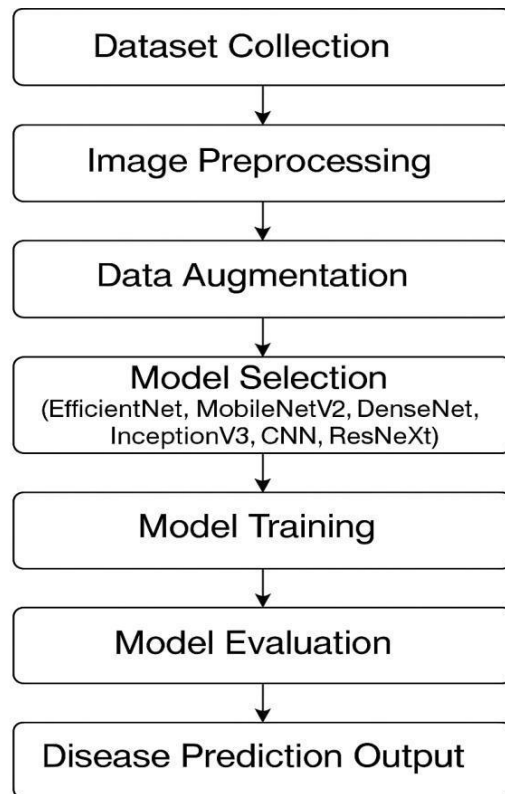
7. Disease Prediction Output

After evaluation, the final model is deployed (locally or via a cloud interface). A user uploads an image of a banana leaf, and the model predicts the disease class.

- Predicted Class: One of the five categories.
- Confidence Score: Probability of the prediction.

This output can be extended to trigger further action like suggesting pesticide, informing agronomist, or logging to farm records.

3.2.2 Explanation of Block Diagram



AI-Driven Crop Disease Prediction and Management System

Block-by-Block Explanation:

1. **Dataset Collection:** Real-world images are collected and organized. Ensures a robust and realistic dataset.
2. **Image Preprocessing:** All collected images undergo resizing, noise cleaning, and normalization.
3. **Data Augmentation:** Makes the model resilient by simulating new image scenarios.
4. **Model Selection:** Among several models, ResNeXt is selected based on empirical analysis.
5. **Model Training:** Uses supervised learning with training dataset, optimizing weights through backpropagation.

6. **Model Evaluation:** Measures how well the model performs and generalizes using unseen data.
7. **Disease Prediction:** The final output provides actionable insights to farmers.

This system design ensures a systematic, scalable, and efficient approach to real-time banana disease prediction using cutting-edge deep learning techniques.

3.3 Algorithm

In this project, we use **Deep Learning-based Image Classification** algorithms to automatically detect and classify diseases in banana plants using leaf images. The core algorithmic approach involves Convolutional Neural Networks (CNNs) and pre-trained models like EfficientNet, MobileNetV2, DenseNet, InceptionV3, and ResNeXt. These models form the backbone of our classification pipeline.

Let's understand the algorithmic approach in detail, step by step:

1. Input: Image Dataset

The dataset contains images of banana leaves categorized into:

- Healthy
- Black Sigatoka
- Yellow Sigatoka
- Panama Disease
- Potassium Deficiency

Each image serves as an input sample to the model. The dataset is split as:

- **70% Training Set**
- **15% Validation Set**
- **15% Testing Set**

2. Image Preprocessing

To prepare the images for training, the following preprocessing steps are applied:

- **Resize:** All images are resized to 224x224 pixels for uniformity across models.
- **Normalization:** Pixel values are scaled to a [0,1] range.
- **Tensor Conversion:** Images are converted to PyTorch tensors for model input.

This ensures that all input data is standardized and compatible with deep learning frameworks.

3. Data Augmentation

Since the dataset is real-time and relatively small, augmentation techniques are applied to improve generalization:

- Random rotations
- Horizontal/vertical flips
- Zoom and shifts

This helps reduce overfitting and improves the model's ability to handle unseen data variations.

4. Model Selection

Five different deep learning models were selected and compared:

- **CNN (Custom):** A basic convolutional neural network architecture.
- **EfficientNet:** Known for high accuracy with fewer parameters.
- **MobileNetV2:** Lightweight, mobile-optimized architecture.
- **DenseNet:** Deep network with feature reuse through dense connections.
- **InceptionV3:** Uses factorized convolutions to reduce computation.

- **ResNext** :A Variant of ResNet.

These architectures are evaluated to determine the best-performing model for banana disease prediction.

5. Model Architecture (General Workflow)

Although model architectures vary, most follow this basic flow:

Input Image → Convolutional Layers → Activation (ReLU) → Pooling Layers → Fully Connected Layers → Softmax → Predicted Class

- **Convolutional Layers:** Extract features from the image.
- **ReLU Activation:** Introduces non-linearity.
- **Pooling:** Reduces spatial dimensions, keeps key info.
- **Fully Connected Layers:** Combine extracted features to make a prediction.
- **Softmax Layer:** Outputs probabilities for each disease class.

6. Training

The model is trained using:

- **Cross Entropy Loss:** For multi-class classification.
- **Adam Optimizer:** Efficient parameter optimization.
- **Learning Rate:** 0.0001 (tuned via validation)
- **Epochs:** 10
- **Batch Size:** 32

The model learns by minimizing the loss between predicted and actual labels.

7. Validation and Testing

- After every epoch, the model is evaluated on the validation set to tune hyperparameters.
- Final performance is measured on the test set to ensure unbiased evaluation.

Metrics used:

- Accuracy
- Precision, Recall, F1-Score
- Confusion Matrix

8. Prediction

After training, the model is used to classify new banana leaf images. It outputs the predicted disease class along with the confidence score.